

Interactive robotics for human impedance estimation in a rhythmic task

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Abstract—This paper presents an experimental robotic setup dedicated to human arm endpoint impedance measurement in a cyclic task with physical interactions. Understanding human endpoint impedance adaptation during a cyclic task in interaction with the environment can provide insights and inspiration for the design of innovative robot controllers for collaborative robotics both in manufacturing and service contexts. The ball bouncing benchmark task, classical in human movement science, is selected here for its simplicity, yet representativeness of hand-eye synchronization and intermittent physical interactions. For a better control over the environmental conditions, the task is performed in a simulated environment, where the paddle is actuated by the human participant by interacting with an admittance-controlled collaborative robotic arm. First experimental feasibility and identification results are presented.

I. INTRODUCTION

In the longtime research topic related to the understanding and modeling of human motor control [1], recent developments benefit from computational approaches and concepts closely related to the automatic control field. The central nervous system (CNS) and the intrinsic bio-mechanical properties of humans limbs allow the generation of coordinated and efficient movements during interaction with their environment [2]. Especially, humans excel in tasks involving physical interactions with environments with a wide range of physical properties, potentially unknown beforehand [3]. Humans are capable of adapting the dynamic properties of their limbs [4] to increase performances during the learning of a new task [5], and reach stable and desired behaviour. Understanding and reproducing some properties of how humans interact with their environment during various tasks is thus relevant for the design of innovative robot control strategies [30].

The mechanical impedance of human limbs extends beyond stiffness and has been studied for decades under the control prism [8], and is still an active research topic [9], [10]. The notion of impedance relates kinematics to forces, allowing the modeling of human movements during physical interactions. Very simple models, such as mass-spring-damper (KBI) or Hill type muscle models, have been shown to describe the human behaviour [7]. An important observation is that these models are not intended to *explain* the complete human motor control, but rather *reproduce* some of its aspects (*eg. Hill model with tendons*). More complete models [12], [24] are required to understand the human CNS.

Human movements can be separated in several categories for better modelling. The differences between rhythmic and discrete movements on the side of human motor control were defined and discussed in [13]. This paper will mostly focus on rhythmic tasks. Elbow joint kinematic and dynamic properties during cyclic task without load were studied in [6]. They were able to connect the maximum frequency with the magnitude of the movement, and then with the arm stiffness, observing both the stiffness and frequency increase together. They proposed a relation between the joint stiffness and the muscular activity, stating in corroboration with previous studies [8], that humans can intentionally control their arm stiffness notably thanks to the co-contraction of agonist and antagonist muscles.

In one hand, the studies on human motor control are essential for better interactions between humans and robots [11], and on another hand they inspire new robotic designs for cobots or to try to reproduce human performances. A bio-inspired torque-stiffness control strategy for dynamic walking was proposed in [14], where central pattern generators (CPG) shape the variations of a robot joint stiffness. This approach was shown to improve the energy efficiency of the task compared to a torque control strategy alone.

In previous work at the intersection of robotics, control and human movement sciences [15], [16], the authors studied the human CNS during cyclic movements in a ball bouncing task in a purely kinematic simulated environment. They were able to reproduce human-like performances using a CPG-based control structure. The Matsuoka oscillator played the role of the CPG, taking the kinematic data of the ball as input to control both amplitude and frequency of the elbow joint angle of a simulated human arm.

This paper proposes a robotic experimental set-up to identify the human impedance by means of an interactive robot. The experience is designed to identify the human arm endpoint impedance variations during a cyclic task, requiring eye-hand coordination, *i.e.* the ball bouncing task. Compared to the previous works [15], [16] on the same task, the modeling objective thus goes beyond the kinematic level, and concentrates now on the interaction forces. Studying the behaviour of the human arm during this kind of tasks could yield new robust strategies for robotics.

The rhythmic ball bouncing task and human impedance model are recalled in Section II. How interactive robotics may be used to identify human impedance is then detailed in Section III. The experimental implementation details are given in Section IV, and preliminary results using this implementation are presented in Section V.

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II. TASK DESCRIPTION AND IMPEDANCE ESTIMATION

In this section, the modeling of the ball bouncing task and the assumptions regarding the human impedance during a rhythmic task are briefly recalled.

A. The ball bouncing task

1) *Ball kinematics*: The ball bouncing task is a well known task in Human Movement Science, in which subjects, often experts, are asked to bounce a ball to reach a target height h , either using their hand or a paddle. The task is depicted on Fig. 1.

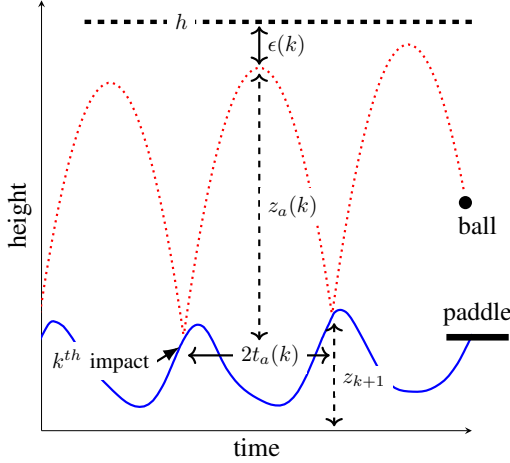


Fig. 1: Paddle (solid blue line) and ball (red dotted line) trajectories during the ball bouncing task with target height h (black dashed line).

Between two impacts k and $k+1$ at times t_k and t_{k+1} , the motion of the ball is described by the ballistic equation (1), with $z(t)$ the ball position, $\dot{z}_k^+$ the post-impact velocity of the ball at t_k , z_k the ball impact position at t_k , t the time elapsed since the last impact k and g the gravitational acceleration.

$$z(t) = -\frac{1}{2}gt^2 + \dot{z}_k^+t + z_k \quad (1)$$

The impact of the ball on the paddle is modeled by (2), using a restitution coefficient α in a simplified manner, as described in [15]. Paddle position is denoted p , m_b and m_p are respectively the ball and the paddle masses, and $\dot{z}_k^- = \dot{z}_{k-1}^+ - gt$ is the velocity of the ball just before the impact k .

$$\dot{z}_k^+ = \max\left(\dot{p}_k, \frac{m_p}{m_p + m_b}((1 + \alpha)\dot{p}_k - \alpha\dot{z}_k^-)\right) \quad (2)$$

2) *Ball bouncing feasibility domain*: Several characteristic constants can be derived from the previous equations. The time of apex between impacts k and $k+1$ is $t_a = \dot{z}_k^+/g$, and the frequency of the task $f = 1/(2t_a)$, that is $f = g/(2\dot{z}_k^+)$, allowing to link the ball apex height z_a , the post-impact velocity and the frequency:

$$z_a - z_k = \frac{(\dot{z}_k^+)^2}{2g} = \frac{\dot{z}_k^+}{4f} \quad (3)$$

Results found in [6] pointed out that for an amplitude of cyclic movements of the elbow joint, around 1.25rad, the

maximum frequency was 2Hz, growing up until 8.4Hz for almost null amplitudes. Since the ball bouncing task requires an active synchronization between the movements of both the arm and ball, smaller frequencies and magnitudes should be considered in order to avoid designing an almost impossible task, or a task requiring complex arm motions. However, this paper focuses on rhythmic and not discrete movements (differences between which are discussed in [13]), therefore the task frequency should be high enough to yield a rhythmic behaviour. The overall objective being to reproduce this rhythmic task using interactive robotics, we provide below a set of feasible conditions for this task deduced from the previous equations and post-impact velocities similar to those considered in [31]. For $\alpha = 0.6$, $g = 9.81 \text{ m.s}^{-2}$, $\dot{z}_k^+ = 3.5 \text{ m.s}^{-1}$, $z_k = 0$ and frequencies ranging from 1 to 2Hz, feasible z_a and thus reachable target height h would be comprised in the interval $[0.5; 1]\text{m}$.

B. Human impedance in a rhythmic task

In a cyclic task performed by a human arm, its endpoint impedance defines the dynamic behavior of the arm when it is deviated by an external perturbation from the nominal trajectory which it would follow without the perturbation, called virtual trajectory in [17]. The KBI model (4) for the mechanical impedance relating forces to motion is considered in this work:

$$\delta f = I\delta\ddot{x} + B\delta\dot{x} + K\delta x, \quad (4)$$

where δx is the difference between the virtual and the perturbed trajectory of the human hand and δf the difference between the virtual and perturbed interaction force. K , B and I are respectively the stiffness (in N.m^{-1}), damping (in N.s.m^{-1}) and mass (in kg) $\mathbb{R}^{n \times n}$ matrices, n being the number of Cartesian DOF.

The introduction of perturbations is thus necessary for impedance identification. Perturbations can have various durations, but for a task in movement, they usually last few hundreds of milliseconds [18], [19], [25], [29], and can be injected either as forces/torques or positions perturbations. The magnitude of the perturbations remains low, since most common impedance models of human limbs are linearizations [26], that hold for small movements δx . Typical force perturbations range around 4N [19], [25], [28], or for position perturbations, below 1cm [18].

This paper does not intend to differentiate the contribution of reflexes from intrinsic properties of the human arm, like it was done in [24], but rather try to observe the global behaviour of the arm in the same way as [25], for rhythmic tasks. To that purpose, the simplified KBI model (4) for the human arm behaviour is proposed at the endpoint level, without coupling between the different axes, that is with diagonal parameter matrices. Considering that the ball bouncing task is reduced to the vertical z -axis, the identified impedance reduces here to a single degree of freedom:

$$\delta f_z = I_z\delta\ddot{z} + B_z\delta\dot{z} + K_z\delta z, \quad (5)$$

Moreover, since this model cannot account for voluntary action, the latter should be discriminated by experimental design and data post-treatment (see Section V-B).

III. INTERACTIVE ROBOTICS AS A TOOL FOR HUMAN IMPEDANCE ESTIMATION

This section presents the proposed methodology for the use of interactive robotics to help the analysis of human impedance during rhythmic tasks. The overall interactive system is presented, the robot control scheme is described, and the impedance estimation method is detailed.

A. Methodology

The ball bouncing task takes place in a simulated environment in which experimental conditions (gravity, restitution coefficient...) are easier to control. The paddle vertical position is connected to the robot's endpoint position along the z axis (Fig. 2). The robot is operated by the human thanks to a handle connected to its end-effector with an interface including a force/torque (f/t) sensor (see Section IV-A for further details). A scaling factor can be applied between real robot movement and the simulated environment to adjust the simulated kinematics.

As shown in Fig. 2, the simulated environment is projected on a screen seen by the human operator and comprises: i) the paddle, only moving along the vertical axis, ii) the ball, also locked in a 1 DOF movement, iii) the target height h , shown by a red dashed line. The goal is to reach a cyclic behaviour on the human side, while bouncing the ball at the given target height in the simulated environment. Since the considered bouncing task is one dimensional, a planar 3 DOF robot evolving in the vertical plane (Oxz) is sufficient for the task.

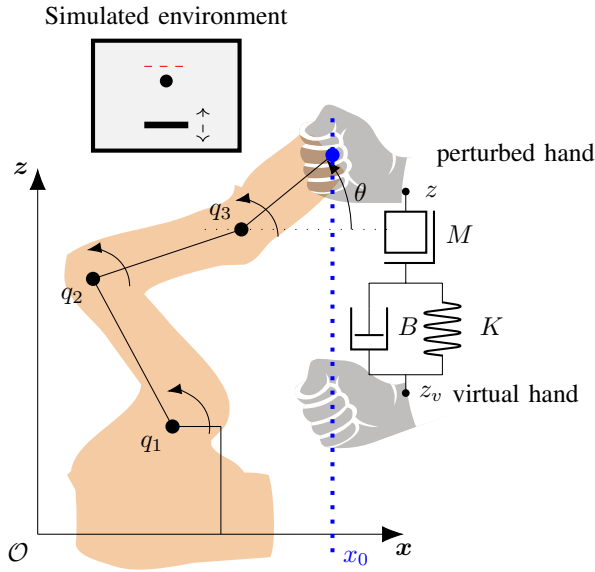


Fig. 2: Proposed interactive system for impedance estimation: a human interacts with the 3 DOF robot thanks to a handle. q_i are the joint positions of the robot, θ is the endpoint orientation, z is the endpoint position along the z axis, represented here after a perturbation responsible for the hand deviation from its nominal trajectory denoted by z_v , the virtual position.

B. Interactive robot control

The complete control scheme is shown in Fig. 3. The robot block represents the robot arm dynamics with its inner joint currents and velocities loops. The control inputs to the robot are therefore joint reference velocities $\dot{q}_d$. The represented controller is composed of three control loops described below.

a) *Main control loop:* With respect to the ball bouncing task, the robot's apparent dynamics should be minimized to be as transparent as possible to the user. To this end, the main control loop realizes a Cartesian admittance control along the z -axis based on the interaction forces f_e , following a similar methodology as in [20], [21]. Namely, C_z is a single-input single-output (SISO) proportional-integral (PI) controller with gains K_p and K_i . Its input is the z component of the sensed interaction force f_z and its output is a reference Cartesian speed of the robot's end-effector $\dot{z}_d$. The Cartesian speed reference is transformed into joint speed reference through the inverse Jacobian $\hat{J}(q)^{-1} \in \mathbb{R}^{3 \times 3}$, which is well defined in the particular case considered here, since the robot is maintained outside its singular configurations by a secondary joint-space position controller. The tuning of the admittance control parameters is presented in Section IV-B.

b) *Secondary control loop about x_0 :* Since the studied ball bouncing task only occurs on the z -axis, the robot motion along the x -axis should be minimized. To avoid the coupling effects in Cartesian space as much as possible, the main admittance controlled is realized at the endpoint level in the Cartesian space, only using the force along the z -axis. However, errors between the estimated and real Jacobian, as well as tracking errors in the velocity loops may generate small but undesired movements along the x -axis. To alleviate this effect, an additional control loop is used to maintain the robot endpoint position about a constant x_0 position, using a Cartesian proportional control of gain K_x .

c) *Secondary control loop about q_0 :* A last sub-task is implemented to avoid unforeseen joint reversal and provide the robot a more predictable movement. The related secondary controller is therefore a simple joint position proportional control of gain K_2 about a given configuration q_0 , which generates an additional joint speed reference $\dot{q}_2$. Since this task should not interfere with the main one along the z -axis, it is projected to the main task's null space using the methodology described in [22], [23], and exploiting the fact that a 3 DOF manipulator is redundant for the realization of the main task. The Jacobian of the main task J_1 is equivalent to the second row of the global Jacobian $J \in \mathbb{R}^{3 \times 3}$. The projected joint speed reference $\dot{q}_2^p$ obtained by projection as shown in (6), is designed to have no effect on the z -axis motion.

$$\dot{q}_2 = K_2 (q_0 - q) \quad (6)$$

$$\dot{q}_2^p = N_2(q) \dot{q}_2 \quad (7)$$

$$N_2(q) = I - J_1(q)^W J_1(q) \quad (8)$$

where the pseudo-inverse A^W of a matrix A is defined in (9), with W a weight matrix that can be the identity matrix

for low dynamics, or the inertia matrix of the robot to ensure the projection in the null-space for movements with greater accelerations [22]. Here the identity matrix is used $\mathbf{W} = \mathbf{I}_3$.

$$\mathbf{A}^{\mathbf{W}+} = \mathbf{W}^{-1} \mathbf{A}^t (\mathbf{A} \mathbf{W}^{-1} \mathbf{A}^t)^{-1} \quad (9)$$

These three combined controllers provide the robot arm with the behaviour of an inertia sliding on a rail, along a vertical line.

C. Human impedance estimation

1) *Perturbation inputs:* To estimate the arm endpoint impedance of a human during a co-manipulation, as described in Section II-B, perturbations need to be introduced in the task. In order to obtain brief perturbations (< 50ms), the inner velocity control loop is bypassed and the robot is torque-controlled during the perturbation window. In this time frame, the outer-loop control of the robot is considered constant, since it has a response time longer than 50ms. Therefore, the last torque input τ_0 before the perturbation is added to the perturbation torque corresponding to an endpoint force $\mathbf{f}_p$ and transformed to the joint level using the transposed Jacobian. The resulting input torque for the robot inner loop is given in (10).

$$\tau = \mathcal{J}(q)^t \mathbf{f}_p + \tau_0 \quad (10)$$

For the considered task, $\mathbf{f}_p$ has only a component along the z -axis. For the impedance identification, these perturbations are introduced at pseudo-random instants of the paddle cycle. In future experiments, the instants of the perturbation will be more finely controlled, but still need to have some randomness to prevent the participants to anticipate the perturbations. Their magnitude is constant in a trial, but their sign is also chosen randomly.

2) *Virtual trajectory and force:* The considered impedance estimation method is based on the fact that the perturbations induce deviations $\delta \mathbf{f}$ and $\delta \mathbf{x}$ (cf (4)) from the *virtual trajectories*, which would have taken place without perturbations. By essence unknown, these virtual trajectories both in hand positions and forces need to be reconstructed. In the literature, the average previous unperturbed trajectories were used in [28], to estimate the virtual trajectory, while [29] fitted the cyclic patterns of previous data to a sinusoidal expression. A prediction algorithm with an unbiased error estimation, lesser than 1cm after 200ms, was proposed in [18]. Since the impedance evaluation is done offline, the trajectory of the arm *after* the perturbation is available. Looking at the arm unperturbed trajectory given in Fig. 4, it can be observed that the movement can be approached using cubic splines. Fig. 4 presents an example of an estimated virtual trajectory against a real perturbed trajectory. More details about this spline-based reconstruction and its influence on the results are given in Section V-B.

3) *Parameters estimation:* Based on (5), the impedance parameter vector $\boldsymbol{\xi} = [K_z \ B_z \ I_z]^T$ can be estimated by linear mean squares, solving the overdetermined system of equations resulting from the evaluation of (11) on a perturbed trajectory. Residuals are denoted ρ .

$$\delta f_z = [\delta x_z \ \delta \dot{x}_z \ \delta \ddot{x}_z] \boldsymbol{\xi} + \rho \quad (11)$$

The identification being performed offline, centered differences can be used to obtain $\delta \dot{x}_z$ and $\delta \ddot{x}_z$.

IV. EXPERIMENTAL IMPLEMENTATION

This section details the specific hardware and software choices that were made for the implementation of the methodology presented in Section III. Obtained results are presented in Section V.

A. Apparatus

The robot used in this study is the 5 DOF KUKATM youBot arm. Low-level joint current and velocity PI control loops are embedded into the motor servo controller modules, with control gains adjusted compared to the factory settings. Higher level control loops (position, force) are implemented using ROS running on a dedicated computer. A rate of approximately 1kHz for the higher level control loops is achieved. The first and the last joints of the 5 DOF KUKA youBot arm are not required for the experiment and are thus maintained at fixed positions through joint position control. On Fig. 2, q_1, q_2, q_3 refer to respectively the 2nd, 3rd and 4th joints of the KUKA youBot arm.

External sensors integrated to this setup are the ATITM Mini45 6 axis f/t sensor and the Optitrack V120:Trio motion capture camera. Measurements from the f/t sensor are acquired at 1kHz, and are used in real time for the admittance control loop (Fig. 2). The camera measurements have a rate of 120Hz and are used in this study only for offline analysis purposes. The acquired mocap position corresponds to the robot handle position held by the human hand.

To compute the interaction forces used in Fig. 3, the f/t sensor measurements are debiased and transformed into the robot base frame, and the effect of gravity on the sensor is compensated. To this end, the mass and arm lever seen by the sensor while it is contact free, as well as the bias are determined using least square optimization. The obtained interaction forces are low-pass filtered at 30Hz using a 4th order Butterworth digital filter, and only the vertical force component f_z is used in closed-loop, as the task is restricted to the z axis.

For the offline analysis and impedance estimation, unfiltered interaction forces are considered and post-treated differently from the control loop. Force and mocap positions are interpolated at 1kHz and low-pass filtered at 25Hz using the same zero-phase fourth order Butterworth filter. These post-treated forces and mocap positions correspond to the notations f_z and x_z in (5).

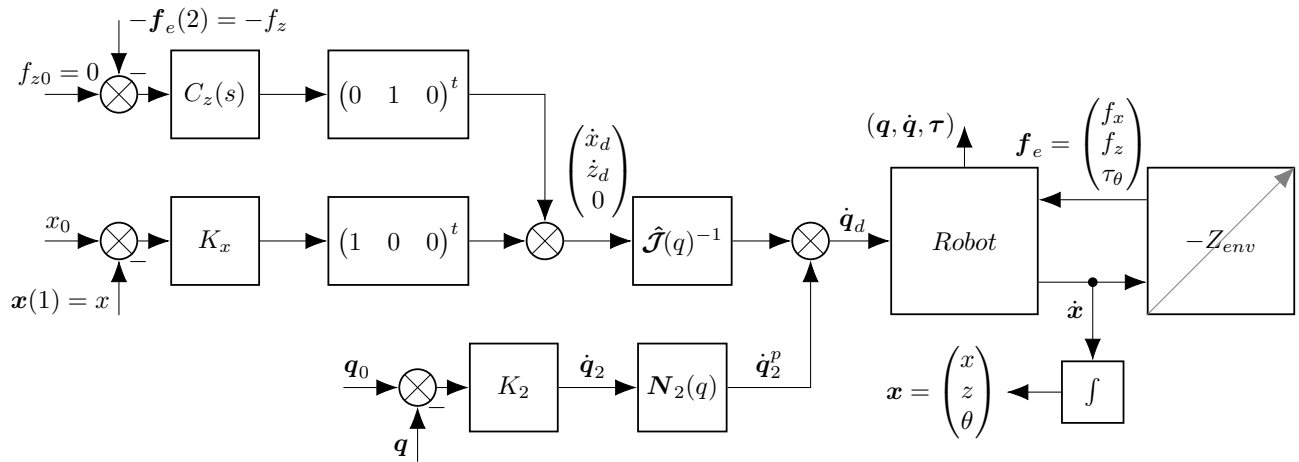


Fig. 3: Block diagram of the interactive robot control for a guided task on a 1 dimensional trajectory along the z axis. $\mathbf{f}_e$ represents the interaction forces in the robot base frame, $\mathbf{q}$ the robot joint positions, $\boldsymbol{\tau}$ the joint torques and $\mathbf{x}$ the robot's endpoint Cartesian coordinates. $C_z = K_p + \frac{K_i}{s}$ is a proportional integral (PI) controller responsible for the interactive admittance control.

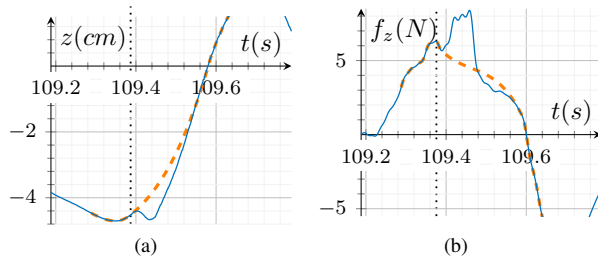


Fig. 4: (a) Hand position and (b) Interaction force, both along the z -axis: realized perturbed trajectories (blue solid lines), versus estimated virtual trajectories (orange dotted ones lines), with the onset time of the perturbation of 30ms duration (vertical black dotted line).

B. Admittance control gains tuning

The admittance control gains are subject to the classical performance-robustness trade-off. To improve transparency and reduce the apparent robot dynamics, higher gains are desired. At the same time, considering that large scale experiments involves several participants, the admittance control loop need to be robust enough to remain stable when in contact with a large range of stiffnesses (*i.e.* participants hand/arm co-contractions), which limits the control gains. It is known that a robot inertia cannot be completely compensated, and its reduction is limited by the stability [21].

To ensure safe and robust co-manipulation, a stability map was experimentally determined, under two different conditions. The first condition corresponds to a manipulation performed under the worst possible environment, that is, moving the robot in rhythm with intentionally strong co-contraction of both the human hand and arm. The second condition corresponds to a nominal ball bouncing manipulation with lower co-contractions of the arm. In each condition, the resulting closed-loop behavior was sorted according to the following four categories: i) Unstable, impracticable, ii) Critical, barely practicable, iii) Badly damped, practicable but uncomfortable, iv) Well damped, or with no oscillation, comfortable. Conceptual behavior of the closed-loop signals is shown on Fig. 5b. The results of the stability

map are displayed in Fig. 5a, the shaped bullets indicating the tested (K_p, K_i) pairs for the first condition, and the green hatched area indicating the comfortable (iv) category for the second condition. The comfortable stability region highlighted in the second test condition is larger than the co-contracted test condition as it can be expected. More detailed stability analysis is planned in future work.

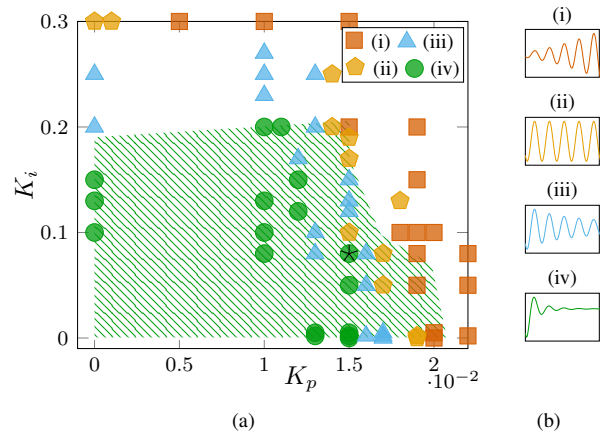


Fig. 5: (a) Experimental stability mapping depending on the admittance control gains. Condition 1: co-manipulation with intentionally high co-contractions of both hand and arm (shapes). Condition 2: co-manipulation with normal co-contractions of both hand and arm during ball bouncing (green hatched area for comfortable manipulation (iv)). The gains used in the experimental set up are indicated with a star. (b) Categories of typical force responses.

V. FIRST EXPERIMENTAL RESULTS

This section presents preliminary results demonstrating the feasibility of the task under the proposed protocol and first insights into impedance identification. More measurements need to be conducted for consolidated and statistically significant results. Future experiments are planned to include several expert participants to achieve longer stable rhythmic movements, with randomized but controlled timing of

the perturbations. A calibration with a known mass-spring-damper system is also to be included.

A. Feasibility of the ball bouncing task

The domain of validity of the task was presented in Section II-A.2. It has been demonstrated feasible with a robot, through two experiments, considering the conditions given in Table I.

f (Hz)	h (m) ^a	g (m.s ⁻²)	$\dot{z}_k^+$ (m.s ⁻¹) ^a	α
[0.9;1.7]	[0.8;1.0]	9.81	[1.32;3.34]	0.6

^aIn the simulated environment.

TABLE I: Tested conditions

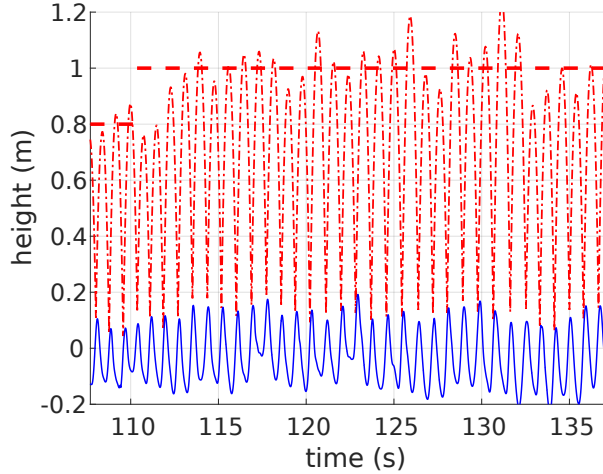


Fig. 6: Ball bouncing task in the simulated environment. The paddle vertical position (solid blue line) reflects the collaborative robot endpoint position. The ball position (dash-dotted red line), and the target height (dashed red line) are depicted. All the data are expressed in the coordinates of the simulated environment.

Fig. 6 shows a part of a bouncing session with perturbations, during a co-manipulation with the robot. The experiment was conducted by the authors themselves. A cyclic behaviour is effectively obtained. Over 30 bounces, the ball apex could be maintained at a virtual height of 1 m within a 6.7% RMS error during the steady state. This variability can be explained by perturbations occurring during the cycles. Indeed, when only the apexes happening after a perturbation are taken into account, the RMS error rises to 7.2%. It is important to underline that a novice participant will need to perform several training experiments to be able to reach the expert level. Participants already practicing juggling tasks are more comfortable with the proposed protocol starting from the first trials.

B. Impedance estimation - first results

1) *Time window selection for estimation:* As mentioned in [25], and reminded in Section II-B, voluntary reactions need to be discarded, because they cannot be properly modeled by (4). Voluntary reactions response time is typically of 100ms. Yet, at a 1kHz rate, [25] found that using only 100ms of data after the perturbation did not bring satisfying results. Indeed to compute the impedance parameters with a

sufficient consistence, enough data are required. Therefore, they assumed that voluntary actions were not dominant in the first 250ms, and used that time window for the evaluation. In the present work, a time window T_{estim} of 200ms was selected for the estimation, based on the literature, and on the direct observation that this time window concentrated the main perturbation effects during the task (see Fig. 4).

In these preliminary experiments, perturbations were induced at randomly chosen instants. For meaningful comparisons and averaging of the estimated parameters, only $n_p = 11$ perturbations not too close to paddle position peaks are selected in the presented results.

2) *Virtual trajectories parameters selection:* The virtual trajectories were obtained using cubic spline interpolation in MATLAB. The interpolation was done at 1kHz, to estimate the virtual trajectory on T_{estim} after a perturbation onset t_{pert} , using data immediately before and after the interval of interest to be reconstructed. To evaluate the quality of this reconstruction, it was applied both on perturbed and non-perturbed cycles (see Fig. 7 for an example). An additional parameter, a delay δt between the beginning of the reconstructed interval with respect to t_{pert} was introduced for comparison purposes. This delay parameter was found to have non negligible influence on the parameter estimation results. Especially, for $0 \leq \delta t \leq 5$ ms, negative parameters were obtained with low determination coefficient R^2 (≤ 0.5) for the linear fit. Incrementally increasing δt to 20ms shows, for the perturbed intervals, a convergence of the mean estimated parameters to constant mean values with significantly reduced relative standard deviations and a dramatically improved mean R^2 of 0.96, against no significant changes on non-perturbed intervals where R^2 remains smaller than 0.6.

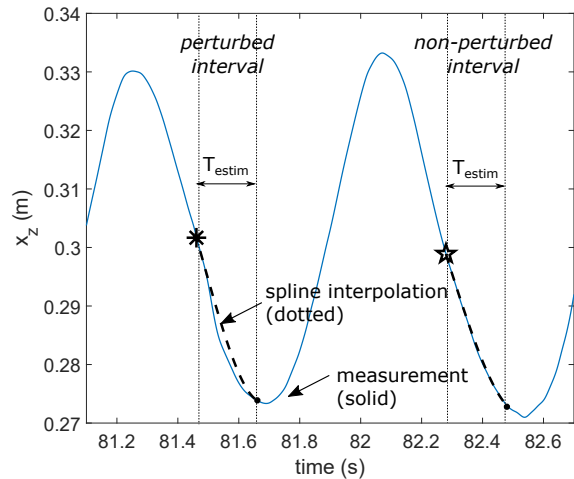


Fig. 7: Example of the mocap position trajectory reconstruction, with and without perturbation.

3) *Parameters estimation:* In [4], [25]–[27] endpoint parameters for the stiffness were found with a magnitude ranging between 40 and 700N.m⁻¹, under static and moving configurations, with and without load for the human arm. Moreover [4], [25], [26] found damping parameters ranging between 8 and 44N.s.m⁻¹, and mass parameters between 0.6

and 2.5kg.

Preliminary results from the authors' trials show most promising results using 200ms for the impedance identification, with $\delta t \geq 15\text{ms}$, possibly to abide by latency in the system. The stiffnesses K_z are found are higher than those mentioned above, with an average of 1337N.m^{-1} and relative standard deviation of 26% on the n_p perturbed intervals. Corresponding average damping B_z and mass I_z coefficients are 3.4N.s.m^{-1} and 0.04kg respectively, however with relative standard deviation slightly higher than 100%. For more significant values, further experiments are to be conducted. Fig. 8 and Fig. 9 show that the reconstructed forces using the identified parameters seem to be able to explain most of the relation between force and movement on perturbed intervals.

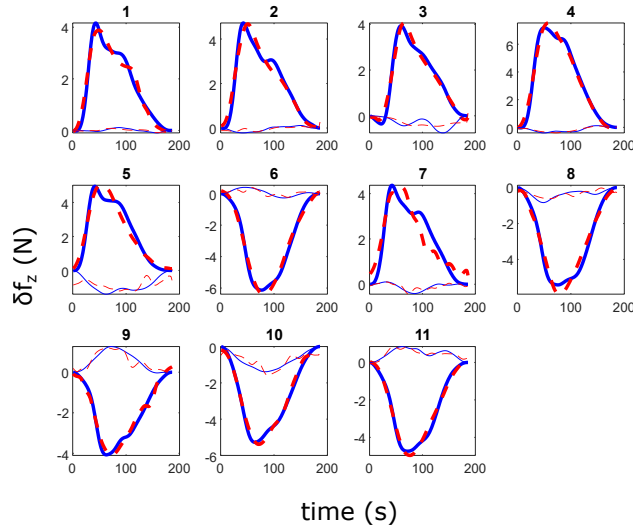


Fig. 8: Force deviations δf_z from the virtual trajectory: comparison between perturbed (bold lines) and non-perturbed (thin lines) cycles over the n_p selected perturbations. δf_z used in identification (solid blue lines) are compared to the reconstructions $\delta \hat{f}_z$ (red dashed lines) based on the identified impedance parameters.

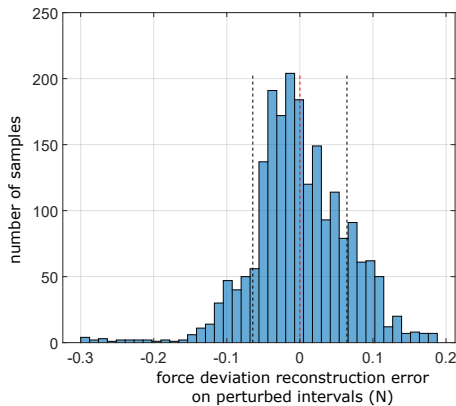


Fig. 9: Reconstruction error $\delta f_z - \delta \hat{f}_z$ distribution for force deviations on perturbed intervals, with the identified impedance parameters.

VI. CONCLUSIONS

The presented first results demonstrate the feasibility of using interactive robotics for human impedance estimation.

Several aspects of the proposed procedure can however be improved. The use of cubic splines for trajectory estimation yields acceptable results for both the positions and velocities, but since cubic splines are C^2 , poorly estimated accelerations were obtained since they were approached with affine functions. Using similar trajectory reconstruction techniques as those described in [18] could improve both virtual trajectory and force estimations. The virtual force estimation is another potential source of errors. Introducing longer perturbations in the outer control loop, directly as a reference f_{z0} in Fig. 3, would simplify the force deviation estimation and could bring more realistic parameters. Other techniques than the presented least squares could be considered to improve the KBI model identification, for example based on linear dynamic system identification methods.

Finally, larger-scale data acquisition on a statistically significant number of experiments with several participants is expected to provide more in-depth insights on the possible cyclic variations of the human endpoint impedance, during a task with physical interaction. Understanding these variations could be used for bio-mimetic rhythmic control design for collaborative robots in interaction with their environment.

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